

Problem 1. Consider the additive model $Y = X^2 + \varepsilon$, where $\varepsilon \sim \text{Unif}[-1, 1]$, i.e., a uniform random variable in the range $[-1, 1]$. Answer the following questions:

1. Plot the conditional distributions $f_{Y|X}(y, 1/2)$ (i.e., the random variable X is equal to $1/2$).
2. Compute and plot the regression function $f(x) = E(Y | X = x)$ and the conditional variance $\text{Var}(Y | X = x)$.
3. Consider the estimator $\hat{Y}(X) = X^3$. Compute $E[(Y - \hat{Y})^2 | X = x]$
4. Indicate the reducible errors for the estimator in sub part (3). Can you find a different estimator with a smaller reducible error?

Problem 2. For each of sub parts (1) through (4), indicate whether we would generally expect the performance of a **flexible** statistical learning method to be better or worse than an **inflexible** method. Justify your answer.

1. The sample size N is extremely large, and the number of predictors p is small.
2. The number of predictors p is extremely large, and the number of observations n is small.
3. The relationship between the predictors and response is highly non-linear.
4. The variance of the error terms, i.e. $\sigma^2 = \text{Var}(\varepsilon)$, is extremely high.

Problem 3. You will now think of some real-life applications for statistical learning. (This question is open-ended, and many solutions are possible.)

1. Describe three real-life applications in which classification might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction? Explain your answer.
2. Describe three real-life applications in which regression might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction? Explain your answer.
3. Describe three real-life applications in which cluster analysis might be useful.

Problem 4. Recall the bias-variance decomposition. (The solution is provided in one of the quizzes.)

1. Provide a sketch of typical (squared) bias, variance, training error, test error, and Bayes (or irreducible) error curves, on a single plot, as we go from less flexible statistical learning methods towards more flexible approaches. The x-axis should represent the amount of flexibility in the method, and the y-axis should represent the values for each curve. There should be five curves. Make sure to label each one.
2. Explain why each of the five curves has the shape displayed in sub part (1).

Problem 5. In this exercise, you shall see the curse of dimensionality in action on a randomly generated dataset. Let's consider a p -dimensional hypersphere of radius 1, denoted by S_p , contained in a p -dimensional hypercube of side length = 2, denoted by C_p , both centered at the origin. The dataset under consideration corresponds to n independent points chosen at random from the uniform probability distribution inside the p -dimensional hypercube.

1. Find expressions for the [volume of the hypercube](#) and the [hypersphere](#) to calculate the expected number of points inside the hypersphere S_p . Since there are n points in the hypercube, what is the fraction of points contained inside the hypersphere? Show a plot of this fraction as a function of p , with p going from 2 to 12. Hint: You might find `math.gamma()` useful to calculate the volume of the sphere.
2. We are now going to verify that these theoretical predictions work in practice using Python. Using the command `numpy.random.uniform()`, generate 100 points uniformly at random in the 2-dimensional hypercube (i.e., C_2 is just a square).
3. Using the above cloud of 100 points, calculate the fraction of points inside S_2 (i.e., the unit circle) with respect to the total number of points inside C_2 .
4. Now repeat (2) and (3) with p going from 2 to 12, and plot the resulting fraction of points as a function of p . What do you see? How does it compare to the theoretical result from (1)? Try repeating with more than 100 samples.
5. Finally, we are going to verify that, as p increases, the distances between pairs of points increase. Using the datasets generated in the previous questions, compute numerically the average pairwise distance among all the points. Plot your result as a function of p . What does the curve look like? How does this relate to the results of (4)?